Answer $z^\star$ · normal → 90m / 6h / 10h

The target cell is read by a frozen copy of the encoder. Each gene question then receives one row.

Target cell

$t \in \{1, 2, 3\}$

90m, 6h, 10h

$x^{\text{tgt}} \in \mathbb{N}^{B \times L_t}$

$L_t \le 512$

EMA encoder

same 3 layers, stop-gradient

$H_{\text{tgt}} \in \mathbb{R}^{B \times L_t \times 768}$

online weights are frozen, so this copy stays put

Gene is on at time t

$z_g^\star = H_{\text{tgt}}[\text{pos}(g)]$

pos(g) is its index in x^{tgt}

the row is in $\mathbb{R}^{768}$

Gene is off at time t

$z_g^\star = a_{\text{null}}$

$a_{\text{null}} \in \mathbb{R}^{768}, \quad \|a_{\text{null}}\| = \sqrt{768}$

one fixed vector, shared by every absent gene

it is a buffer, so it receives no gradient

Stacked answer

$z^\star \in \mathbb{R}^{B \times 128 \times 768}$

one row per gene question

on → row of H_{tgt}

off → a_{null}

The 19 mask questions still receive a $z^\star$, and $\mathcal{L}_{\text{var}}$ and $\mathcal{L}_{\text{abs}}$ skip them.

The cell answer is the ⟨cls⟩ row of H_{tgt}, in $\mathbb{R}^{B \times 768}$, with weight 0.